25901815 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR/IMPROVEMENT/REAPPEARANCE EXAMINATION APRIL 2026

SECOND SEMESTER

Discipline Specific Core Course (DSC) - MG2DSCZGY101 - BIOLOGICAL BASIS OF BEHAVIOR II

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer all questions

Each question carries **1** marks

1. What is the first step in cell communication? [U] / [CO1]

   a). The cell responds to the signal
   b). The cell absorbs nutrients
   c). A signal binds to a receptor
   d). The cell produces energy

2. The process where homologous chromosomes pair up is called ___. [K] / [CO1]

   a). Crossing over
   b). Synapsis
   c). Cytokinesis
   d). Replication

3. G1 and G2 phase of cell cycle are: [K] / [CO1]

   a). Growth phase
   b). Synthesis phase
   c). Lag phase
   d). Termination phase

4. In a monohybrid cross involving a heterozygous pair (Tt × Tt) where T is dominant for tall and t is recessive for short, what is the expected phenotypic ratio in the offspring? [U] / [CO2]

a). 1 tall : 1 short b). 2 tall : 1 short

c). 3 tall : 1 short d). 1 tall : 2 short : 1 tall

5. Which of the following is a sex-linked disorder? [U] / [CO2]

a). Hemophilia b). Sickle cell anemia

c). Albinism d). Tay-Sachs disease

6. The next step after transcription is: [K] / [CO2]

a). DNA replication b). Metaphase

c). Translation d). Anaphase

7. Who discovered the double-helix structure of DNA? [K] / [CO2]

a). Watson and Crick b). Franklin and Wilkins

c). Mendel and Darwin d). Chargaff and Griffith

8. Which of the following is a feature of Cri-du-chat syndrome? [U] / [CO3]

a). Cat-like cry b). Long arms c). Large ears d). High intelligence

9. What is the primary role of genetic counselling? [K] / [CO4]

a). To treat genetic disorders

b). To provide information and guidance to patients and families

c). To conduct laboratory tests

d). To diagnose all genetic conditions

10. Karyotyping is used for [U] / [CO4]

a). DNA sequencing b). Chromosome analysis c). Protein synthesis d). RNA formation

[1x10 = 10]

Part B

Short Answer Questions

Answer any **10** questions

Each question carries **2** marks

11. What is nucleolus? [U] / [CO1]

12. Comment on the role of hormones as signalling molecules. [U] / [CO1]

13. What is the key characteristic of paracrine signaling? [K] / [CO1]

14. What is the phenotypic ratio and genotypic ratio of a dihybrid cross? [U] / [CO2]

15. What is the relevance of DNA replication? [U] / [CO2]

16. How does a test cross help in identifying the genotype of an individual? [K] / [CO2]

17. Explain the key points of chromosome theory of inheritance. [U] / [CO2]

18. What is Turner's syndrome? [U] / [CO3]

19. What is the significance of identifying consanguineous marriages in a pedigree analysis? [An] / [CO4]

20. Name two types of numerical chromosomal aberrations. [K] / [CO3]

21. What is spontaneous mutation? [U] / [CO3]

22. What is eugenics? Give an example. [U] / [CO4]

[2x10 = 20]

Part C

Short Essay Questions

Answer any **5** questions

Each question carries **4** marks

23. Describe the main characteristics of eukaryotic cells? [U] / [CO1]

24. Explain anaphase in mitosis with the help of a diagram. [U] / [CO1]

25. State and explain Mendel's Law of Segregation with an example. [U] / [CO2]

26. Explain the process of transcription [U] / [CO2]

27. Differentiate between phenotype and genotype [U] / [CO2]

28. How can point mutations contribute to genetic diseases? Give an example? [U] / [CO3]

29. Define euthenics and its significance in improving the current human population. [An] / [CO4]

[4x5 = 20]